# Fun With Statistics

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This one is for all the math people out there – I won't speculate on applications, although I'm sure they exist. Also, thanks to Matt Weinberg, who posed the question that got me thinking about this.

Suppose that $\{X_i\}_{i=1}^n$ are independent, non-negative random variables, let $S_k = \sum_{i=1}^k X_i$, and let $Y$ be independent of the $X_i$. One might be interested in calculating $P(S_n \leq Y)$. In general, this is very complex – for example, if $Y$ is deterministic, then we are describing the distribution of $S_n$, which requires taking the convolution of $n$ distributions. However, if $Y$ is exponentially distributed, we have that

$$P(S_n \leq Y) = \Pi_{i=1}^n P(X_i \leq Y). \tag{1}$$

This seems pretty remarkable (at least to me) – there's no obvious reason that these quantities should be related! Nevertheless, it can be proved easily. The idea is as follows: imagine learning whether $S_n \leq Y$ in a series of rounds:

- In round 1, learn $S_1$, and whether $Y \geq S_1$. If so, proceed to round 2.
- In round 2, learn $S_2$, and whether $Y \geq S_2$. If so, proceed to round 3.
- In round 3, learn $S_3$, and whether $Y \geq S_3$. If so, proceed to round 4.
- $\cdots$

Because $Y$ is memoryless (meaning that the distribution of $Y - y$, given $Y \geq y$, is identical for all non-negative $y$), for any realization of $S_{k-1}$ we have

$$P(Y \geq X_k + S_{k-1} | S_{k-1}, Y \geq S_{k-1}) = P(Y \geq X_k). \tag{2}$$

Thus the probability of passing all $n$ rounds is $\Pi_{i=1}^n P(X_k \leq Y)$.

While the argument above is specific to the exponential distribution, it can be generalized. In particular, if $Y$ is light tailed (meaning that $P(Y - y \geq x | Y \geq y)$ is decreasing in $y$ for all $x$) then the equality in (2) becomes an inequality. This yields the following result.

**Proposition 1** *If $Y$ has a light (heavy) tail, then $P(S_n \leq Y)$ is less (greater) than $\Pi_{i=1}^n P(X_i \leq Y)$.*

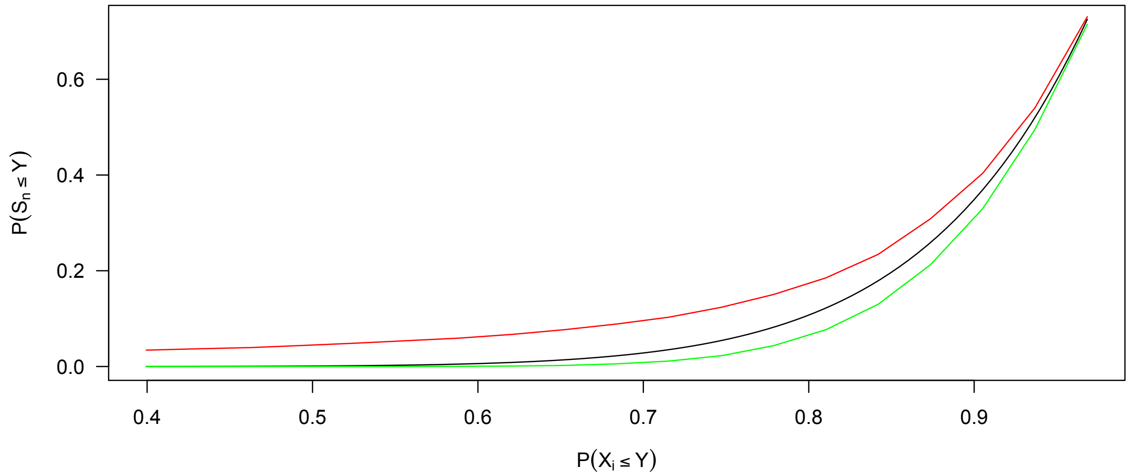


Figure 1: Approximating $P(S_n \leq Y)$ with $\prod_{i=1}^{n} P(X_i \leq Y)$ (black curve). The green curve represents a light-tailed random variable, with $P(Y \leq y) = 1 - e^{-y^2}$. The red curve represents a heavy-tailed random variable, with $P(Y \leq y) = 1 - (1 + x)^{-1/\log(2)}$. It would be interesting to better understand the conditions under which the approximation is and isn't good.

# 1 Code

```
n = 10
N = 10^5

P = rep(1:19)/20
l = rep(0,length(P))
e = rep(0,length(P))
h = rep(0,length(P))
for(i in 1:19){
        p = P[i]
        k = 1
        a = 1/log(2)
        countL = 0
        countE = 0
        countP = 0
        for(j in 1:N){
                x = rbinom(1,n,p) #X_i is bernoulli with probability p
                U = runif(1)
                yL = sqrt(-log(U)) #Light Tailed; F(x) = 1- exp(-x^2)
                yE = -log(U) #Exponential; F(x) = 1- exp(-x)
                yP = U^(-1/a) - 1  #Power Law; F(x) = 1- (1+x)^{-a}
                if(x < yL) countL = countL+1
                if(x < yE) countE = countE+1
                if(x < yP) countP = countP+1
        }
        l[i] = countL/N
        e[i] = countE/N
        h[i] = countP/N
}
x = 1 - P + P*exp(-k)
plot(function(x){return(x^n)},xlim=c(min(x),max(x)),ylim=c(0,max(x)^n),ylab=expression(P(S[n] <= Y
lines(x,l,col='green')
lines(x,e,col='blue')
lines(x,h,col='red')
```